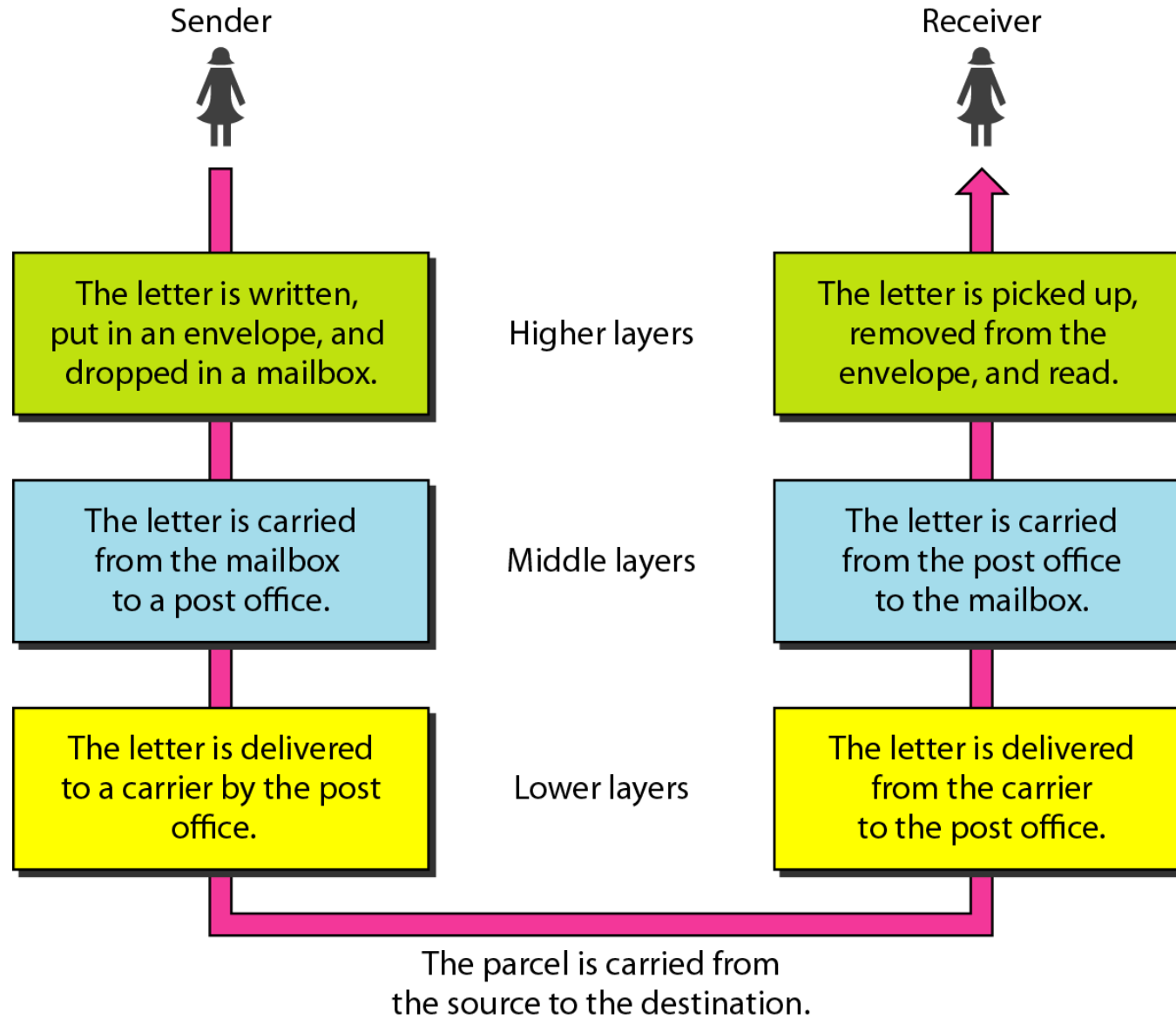


REFERENCE MODEL - OSI

LAYERED TASKS

- *We use the concept of **layers** in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.*



Tasks involved in sending a letter

Layered Tasks

Example: friends communicate through postal mail.

Sender, Receiver, & Carrier

At the Sender Site

Higher layer: sender writes the letter, insert the letter in an envelope, writes the sender & receiver address, & drops the letter in a mailbox.

Middle Layer: letter is picked up by a letter carrier & delivered to the post office.

Lower Layer: letter is sorted at the post office; a carrier transports the letter.

On the way letter go through a central office. Transported by any vehicle.

At the Receiver Site

Lower Layer: carrier transports the letter to the post office.

Middle Layer: letter is sorted & delivered to the recipient's mailbox.

Higher Layer: receiver picks up the letter, opens the envelope, & read it.

Why Layered Architecture is Important in Industry?

- Simplifies network design and development
- Allows independent development of protocols
- Makes troubleshooting faster and systematic
- Supports interoperability between devices from different vendors
- Enables technology upgrades without affecting the complete network
- **Examples**
 - Cisco Router + HP Switch + Dell Server communicate using standard protocols.
 - 5G can replace 4G without changing applications like WhatsApp or Gmail.
 - Cloud providers (AWS, Azure, Google Cloud) use layered architecture for scalable services.

OSI

- The OSI (Open Systems Interconnection) Reference Model was developed by the International Organization for Standardization (ISO) in 1984 to provide a standard framework for communication between heterogeneous computer systems.
 - OSI stands for Open Systems Interconnection. It provides a seven-layer conceptual framework for network communication.
 - The OSI Reference Model consists of seven layers, where each layer performs a specific function and provides services to the layer above it.
- *Note A handy way to remember the seven layers is the sentence "All people seem to need data processing." The beginning letter of each word corresponds to a layer*

Why Do Engineers Still Learn the OSI Model?

- Although the Internet uses TCP/IP, the OSI model is widely used for:
 - Network Troubleshooting
 - Cisco CCNA & CCNP Certification
 - Cybersecurity Analysis
 - Firewall Configuration
 - Packet Analysis using Wireshark
 - Technical Interviews
- **Example**
- If a website is not opening:
 - Check Cable → Physical Layer
 - Check MAC Address → Data Link Layer
 - Check IP Address → Network Layer
 - Check TCP Connection → Transport Layer
 - Check DNS/HTTP → Application Layer
- **Note:** OSI provides a systematic approach to identifying network issues.

OSI Layers in Everyday Applications

- When we send a WhatsApp message, it does not use only one layer.
- It passes through multiple layers:
 - Application Layer → WhatsApp Application
 - Transport Layer → TCP ensures reliable delivery
 - Network Layer → IP finds the destination
 - Data Link Layer → Wi-Fi/Ethernet transfers frames
 - Physical Layer → Radio signals or Fiber transmit bits

Every modern application depends on multiple networking layers working together.

ISO/OSI Reference Model

- Open Systems Interconnection
- The OSI model is primarily a conceptual reference model. Modern networks use the TCP/IP protocol suite, while the OSI model is widely used for learning, troubleshooting, protocol design, and security analysis.
- Provides a standardized framework for developing interoperable networking protocols and services.
- Task of communication broken up into modules or layers
- Value: The reference model defines 7 layers of functions that take place at each end of communication and with each layer adding its own set of special related functions.
- Each layer expects some service from its lower layer, and provides some service to its higher layer.

OSI

Principles for the seven layers

- Layers created for different abstractions
- Each layer performs well-defined function
- Function of layer chosen with definition of international standard protocols in mind
- Minimize information flow across interfaces between boundaries
- Number of layers optimum

Design Issues of the Layers

A number of design issues exist for the layer to layer approach of computer networks. Some of the main design issues are as follows –

Reliability

- Network channels and components may be unreliable, resulting in loss of bits while data transfer. So, an important design issue is to make sure that the information transferred is not distorted.

Scalability

- Networks are continuously evolving. The sizes are continually increasing leading to congestion. Also, when new technologies are applied to the added components, it may lead to incompatibility issues. Hence, the design should be done so that the networks are scalable and can accommodate such additions and alterations.

Design Issues of the Layers

Addressing

- At a particular time, innumerable messages are being transferred between large numbers of computers. So, a naming or addressing system should exist so that each layer can identify the sender and receivers of each message.

Error Control

- Unreliable channels introduce a number of errors in the data streams that are communicated. So, the layers need to agree upon common error detection and error correction methods so as to protect data packets while they are transferred.

Flow Control

- If the rate at which data is produced by the sender is higher than the rate at which data is received by the receiver, there are chances of overflowing the receiver. So, a proper flow control mechanism needs to be implemented.

Design Issues of the Layers

Resource Allocation

- **Computer network** provide services in the form of network resources to the end users. The main design issue is to allocate and deallocate resources to processes. The allocation/deallocation should occur so that minimal interference among the hosts occurs and there is optimal usage of the resources.

Statistical Multiplexing

- It is not feasible to allocate a dedicated path for each message while it is being transferred from the source to the destination. So, the data channel needs to be multiplexed, so as to allocate a fraction of the bandwidth or time to each host.

Design Issues of the Layers

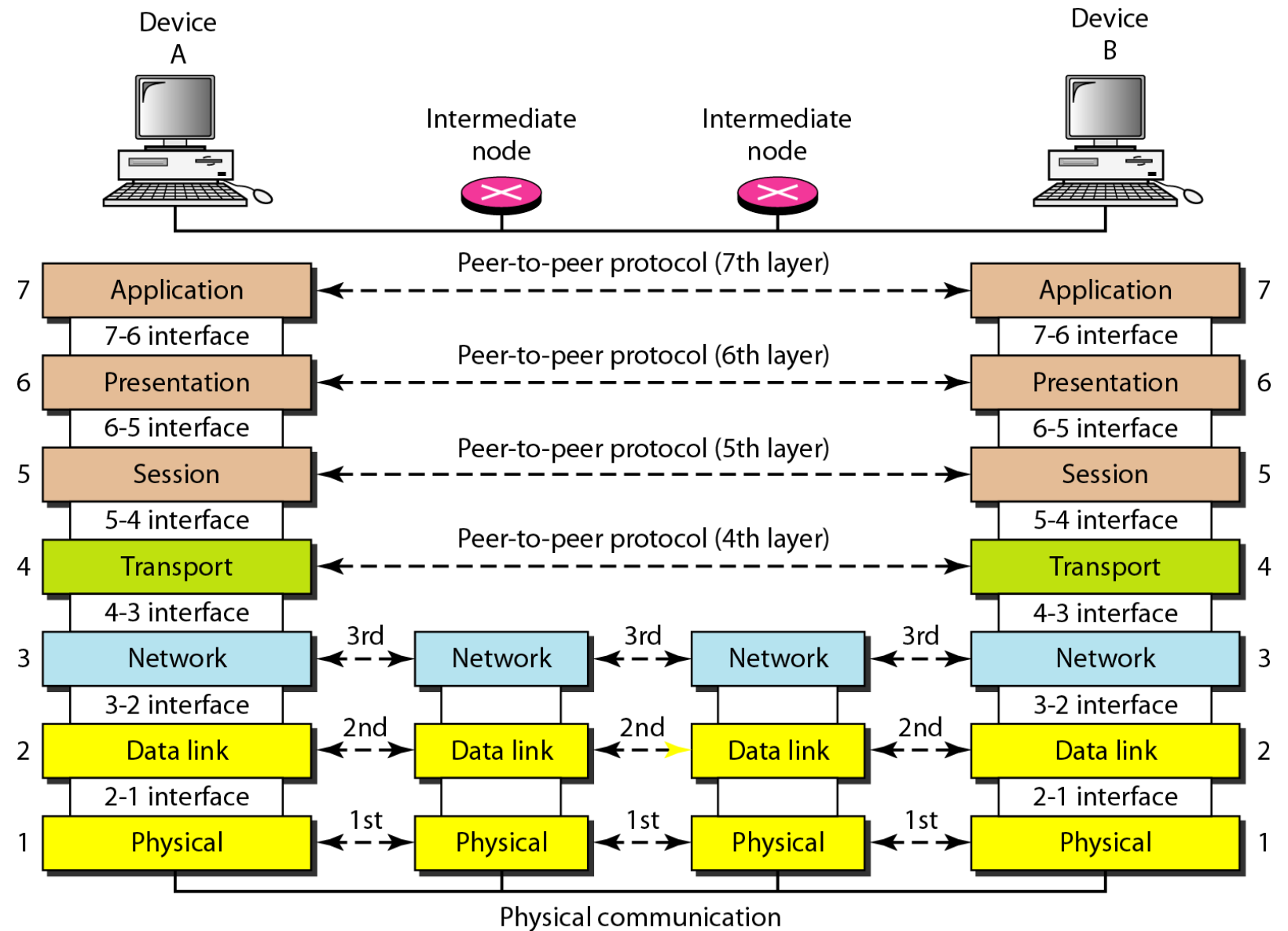
Routing

- There may be multiple paths from the source to the destination. Routing involves choosing an optimal path among all possible paths, in terms of cost and time. There are several routing algorithms that are used in network systems.

Security

- A major factor of data communication is to defend it against threats like eavesdropping and surreptitious alteration of messages. So, there should be adequate mechanisms to prevent unauthorized access to data through authentication and cryptography.

The interaction between layers in the OSI model



File Transfer, Email, Remote Login →

ASCII Text, Sound (syntax layer) →

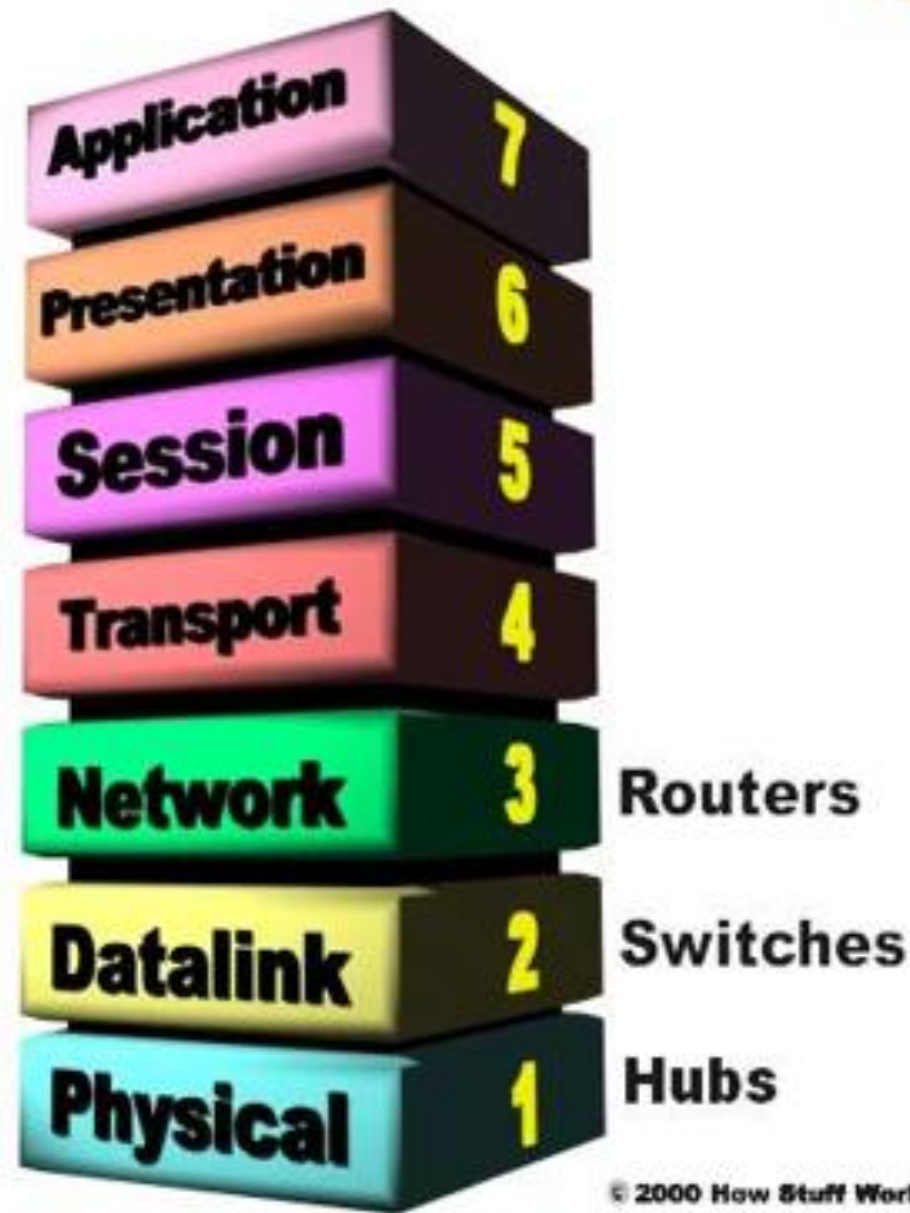
Establish/manage connection →

End-to-end control & error checking
(ensure complete data transfer): TCP →

Routing and Forwarding Address: IP →

Two party communication: Ethernet →

How to transmit signal; coding
Hardware means of sending and
receiving data on a carrier →



Application

Presentation

Session

Transport

Network

Datalink

Physical

Host-to-host

Point-to-point

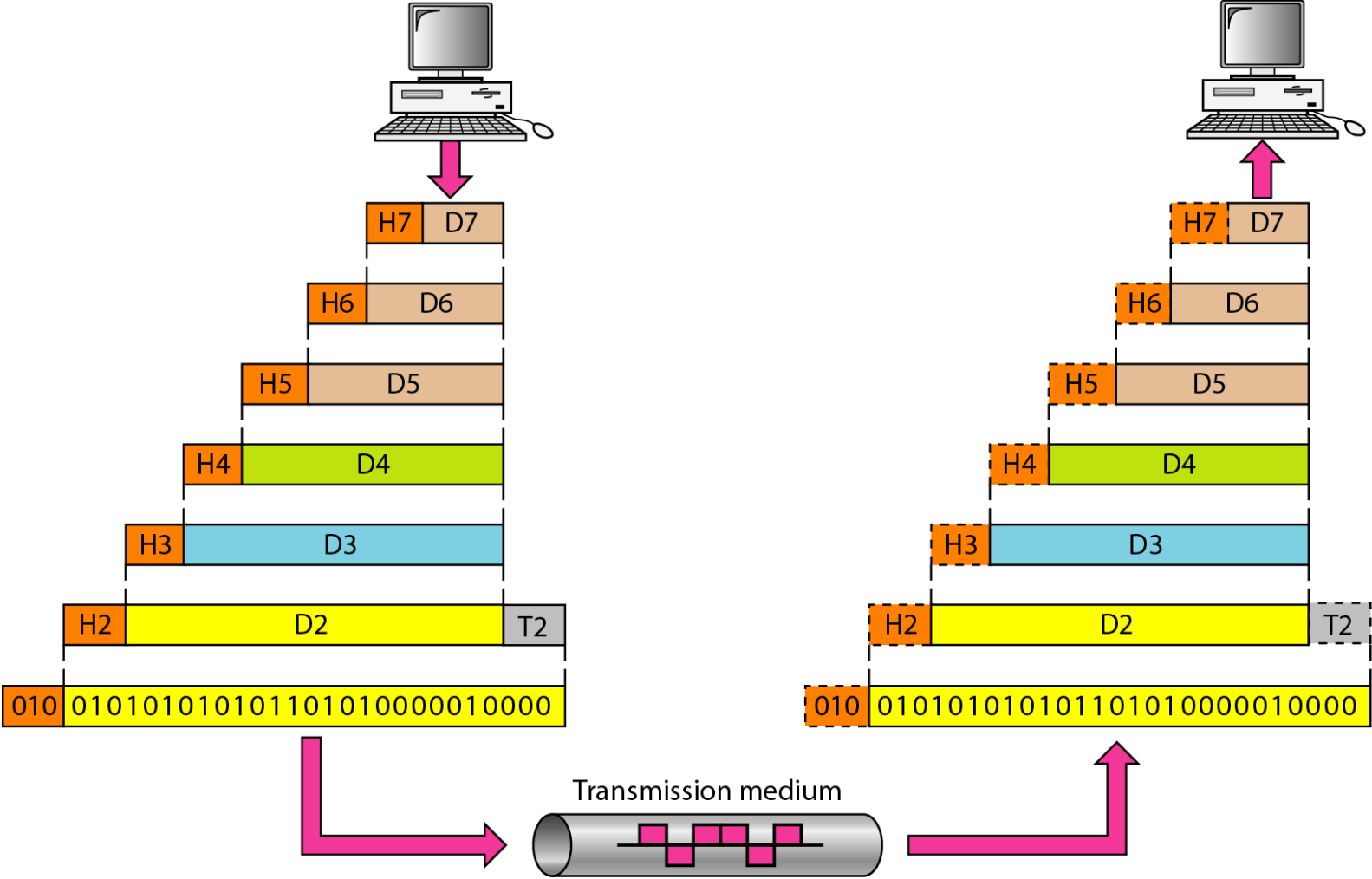
Host-to-host communication

- Host-to-host communication ensures that data is delivered correctly from the source computer (host) to the destination computer (host).
- These layers deal with:
 - User applications
 - Data formatting
 - Session establishment and termination
 - End-to-end reliability
- The communication is between two end devices, regardless of how many routers or switches are in between.

Point-to-Point Communication

- Point-to-point communication means communication between two directly connected devices (called a *hop*).
- Each hop occurs between neighboring devices, such as:
 - PC ↔ Switch
 - Switch ↔ Router
 - Router ↔ Router
 - Router ↔ Switch
 - Switch ↔ PC
- Each device forwards the packet to the next device until it reaches the destination.

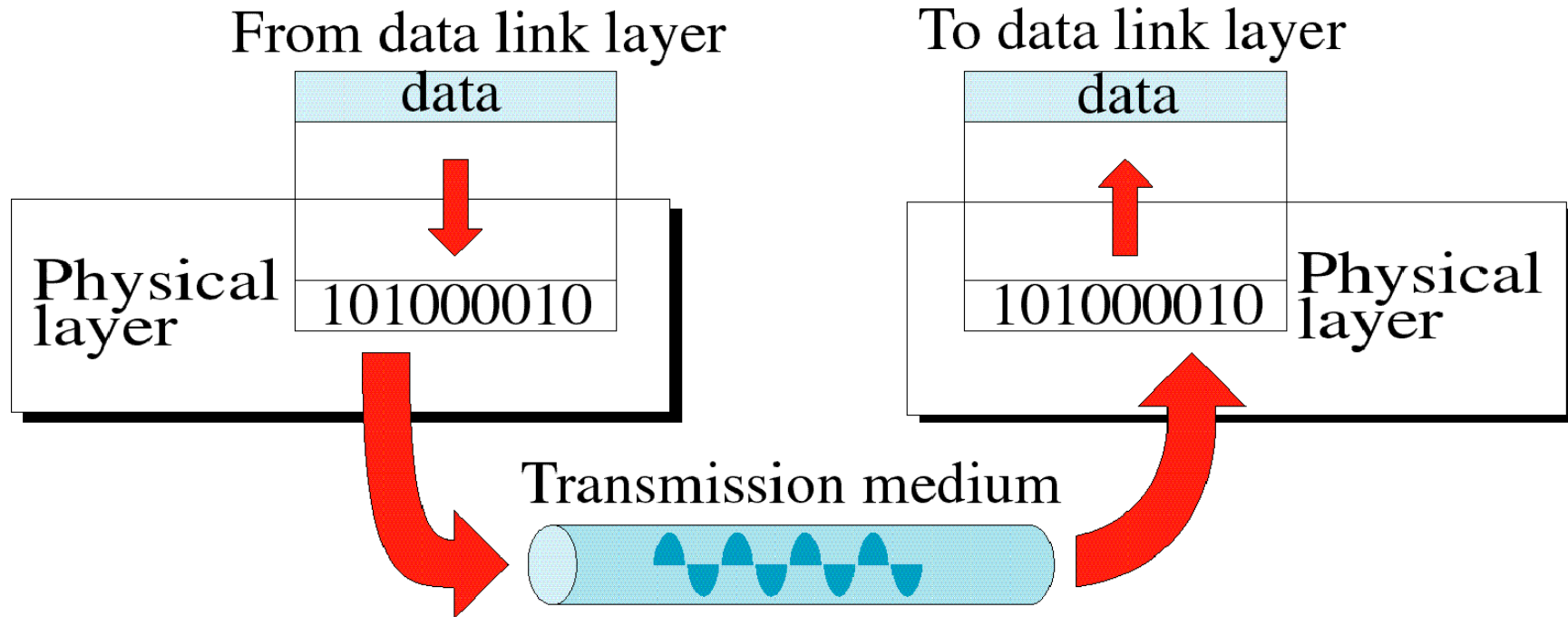
An exchange using the OSI model



Physical Layer

- Transmitting raw bits over a “wire”
 - Make sure a “1” bit is sent as a 1
- The design issues here largely deal with mechanical, electrical , and procedural interfaces, and the physical transmission medium, which lies below the physical layer.
- For example:
 - How many volts represents a “1” or “0”?
 - How long does a bit time last?
 - How many pins does the connector have?
 - How many wires does the transmission media have?
 - Are pulses electrical or optical or waves?

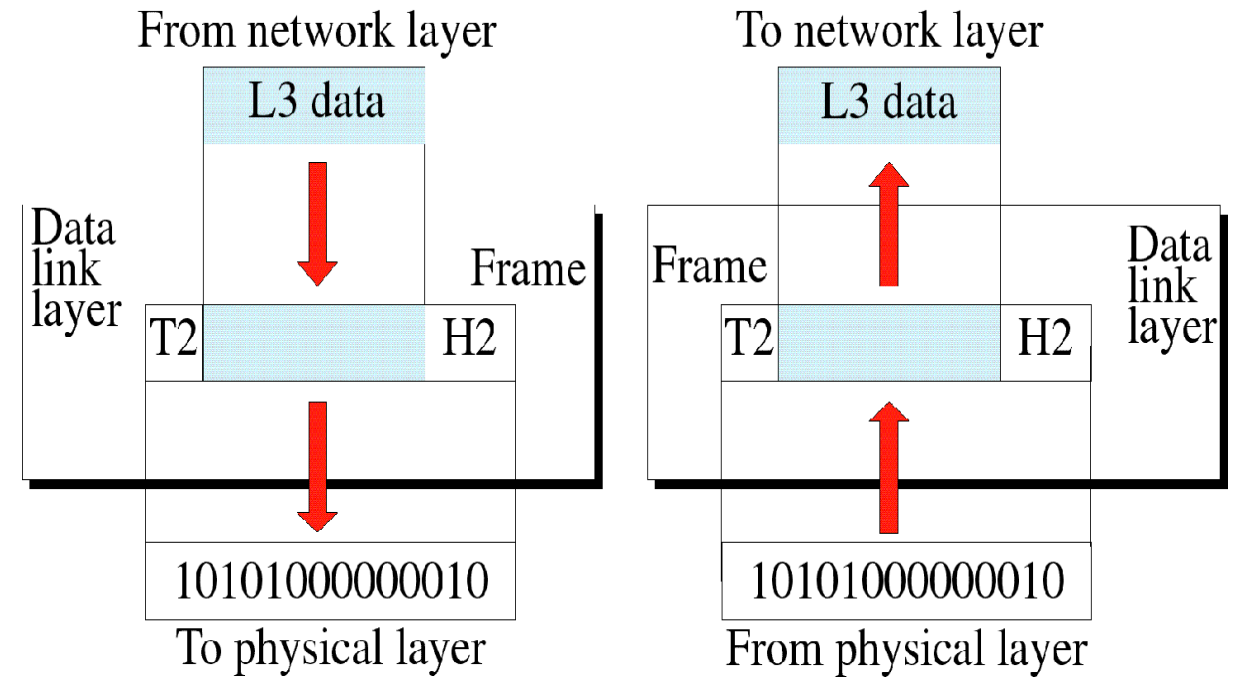
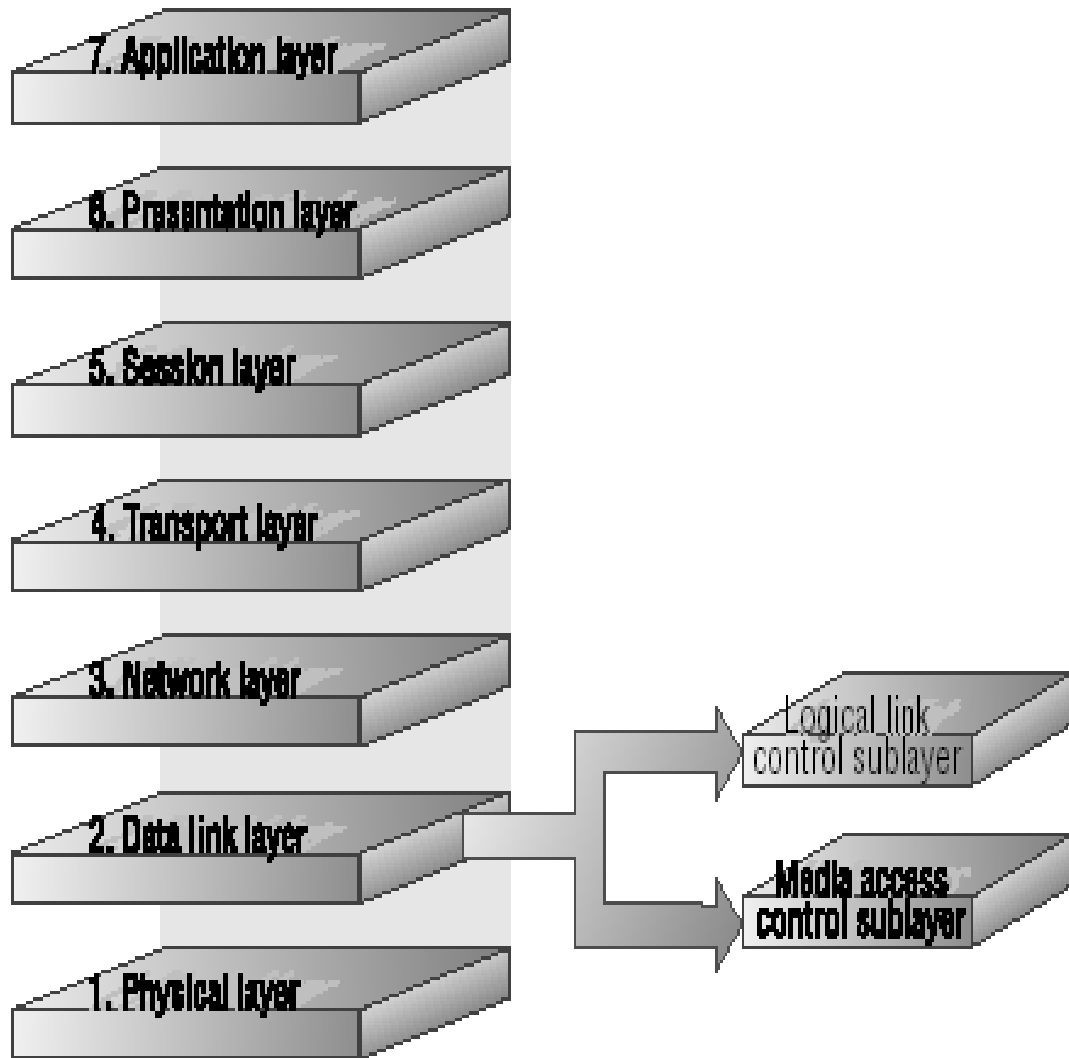
Physical layer



The Physical layer is responsible for moving individual bits from one node to the next.

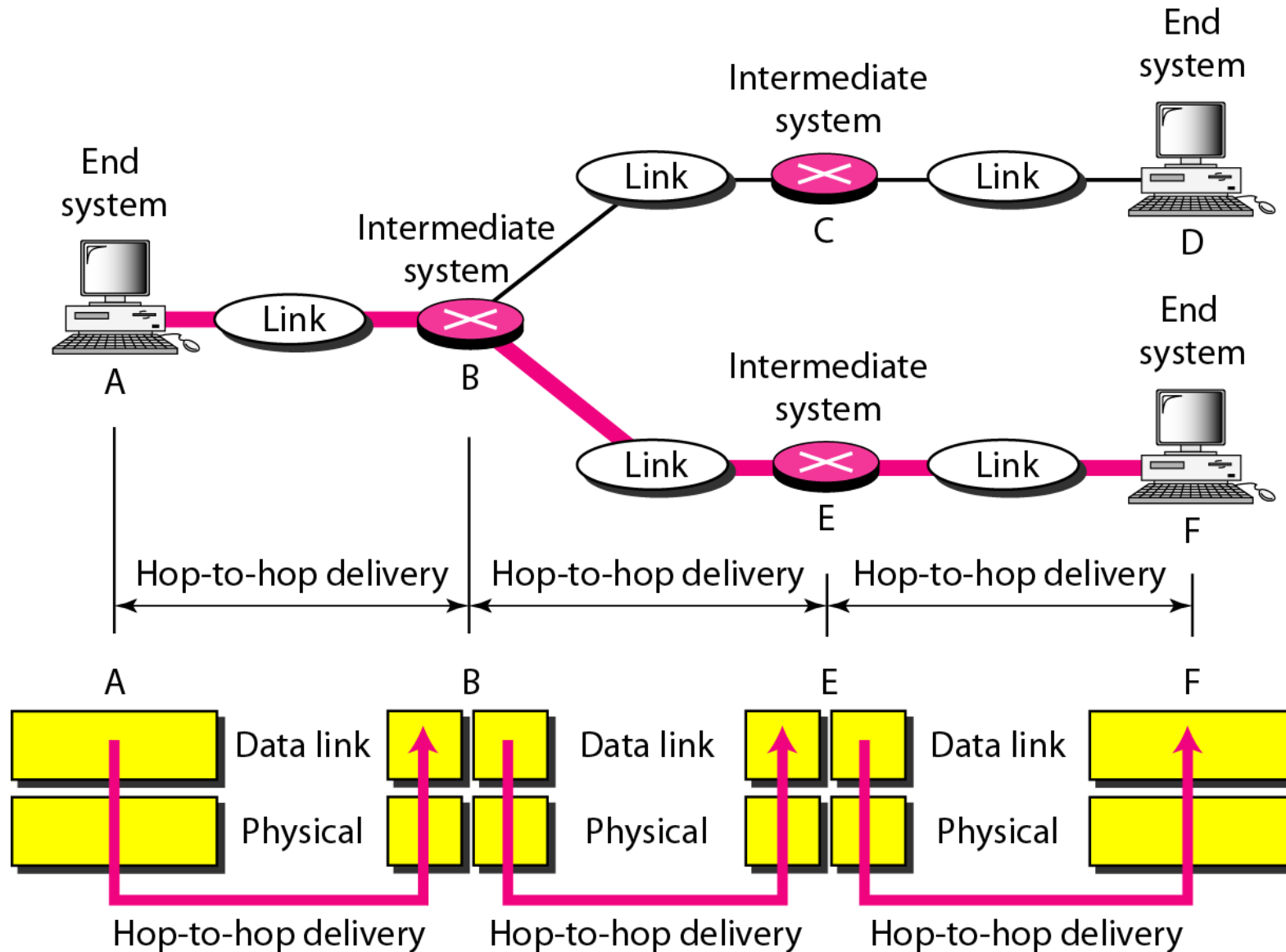
Data Link Layer

- Communication between two machines
- Transforms raw transmission of physical layer into error-free channel.
- Divides physical layer into *frames*
 - messages containing data and control information
- Handles lost, damaged, and duplicate frames.
- Provides flow control between the sender and receiver.
- The IEEE committee felt that the data link layer needed to be defined in greater detail, so they split it into two sub-layers:
 1. **Logical Link Control (LLC)**, which establishes and maintains links between the communicating devices
 2. **Media Access Control (MAC)**, which controls the way multiple devices share the same media channel



The data link layer is responsible for moving frames from one hop (node) to the next.

Hop-to-hop delivery

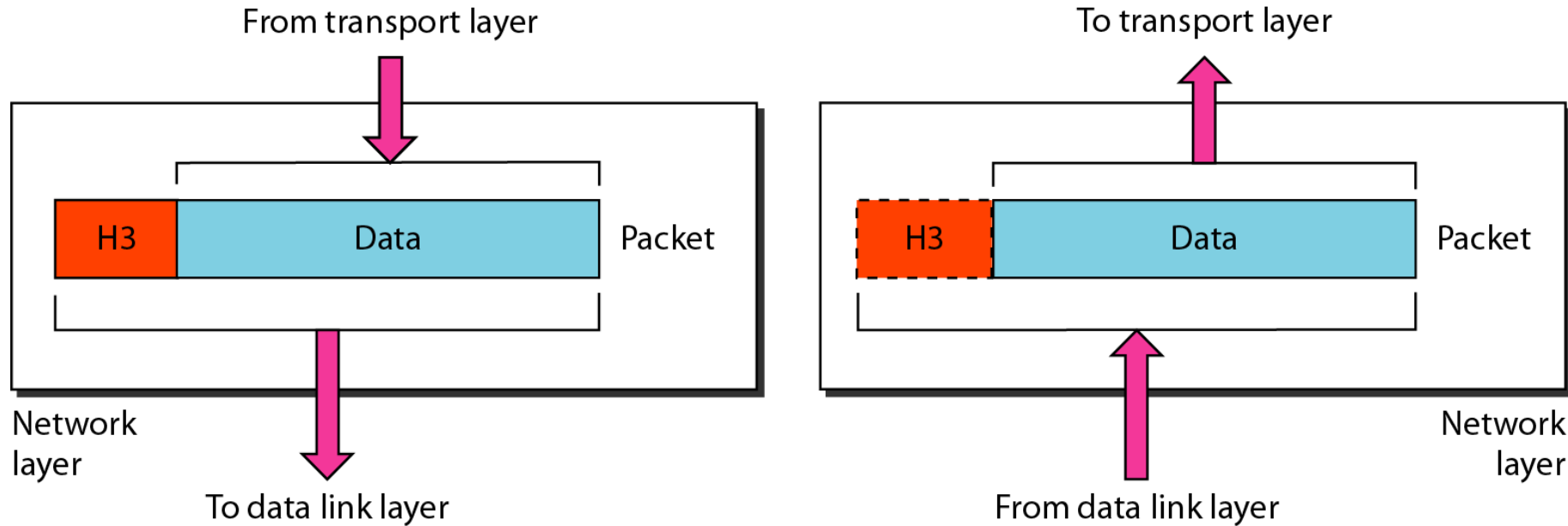


Network Layer

- The network layer makes routing decisions and forwards packets for devices that are farther away than a single link. [Routers are used for routing]
- This layer also determines the quality of service (such as the priority of the message) and the route a message will take if there are several ways a message can get to its destination.
- Handles logical addressing (IP addressing).
- Determines the best route for packet delivery.
- Performs routing and congestion control.

If 2 systems are connected to the same link, there is usually no need for a NL.

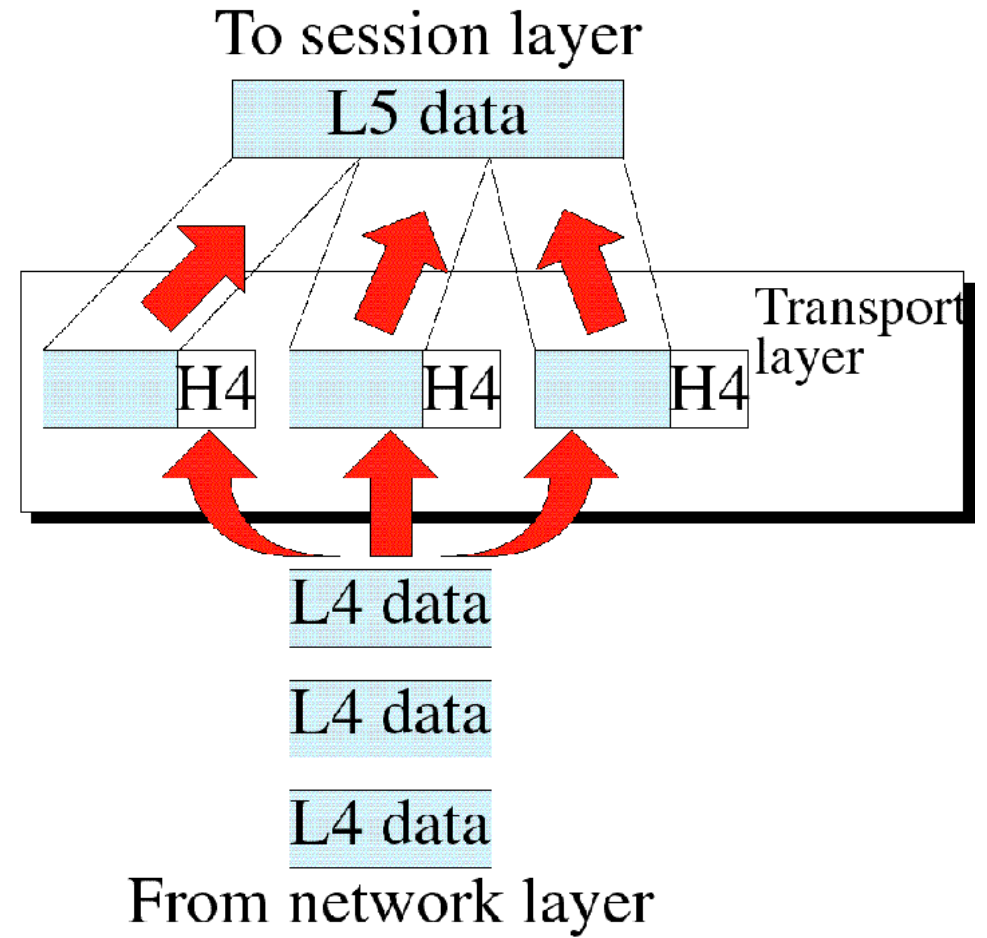
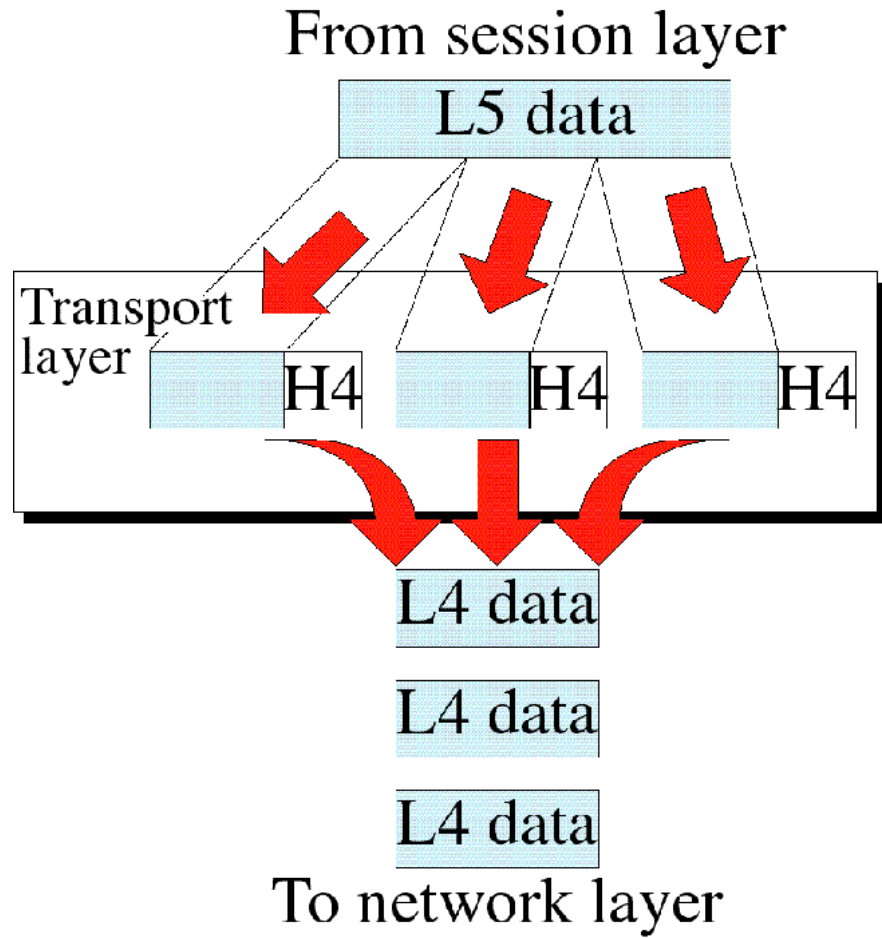
Network layer



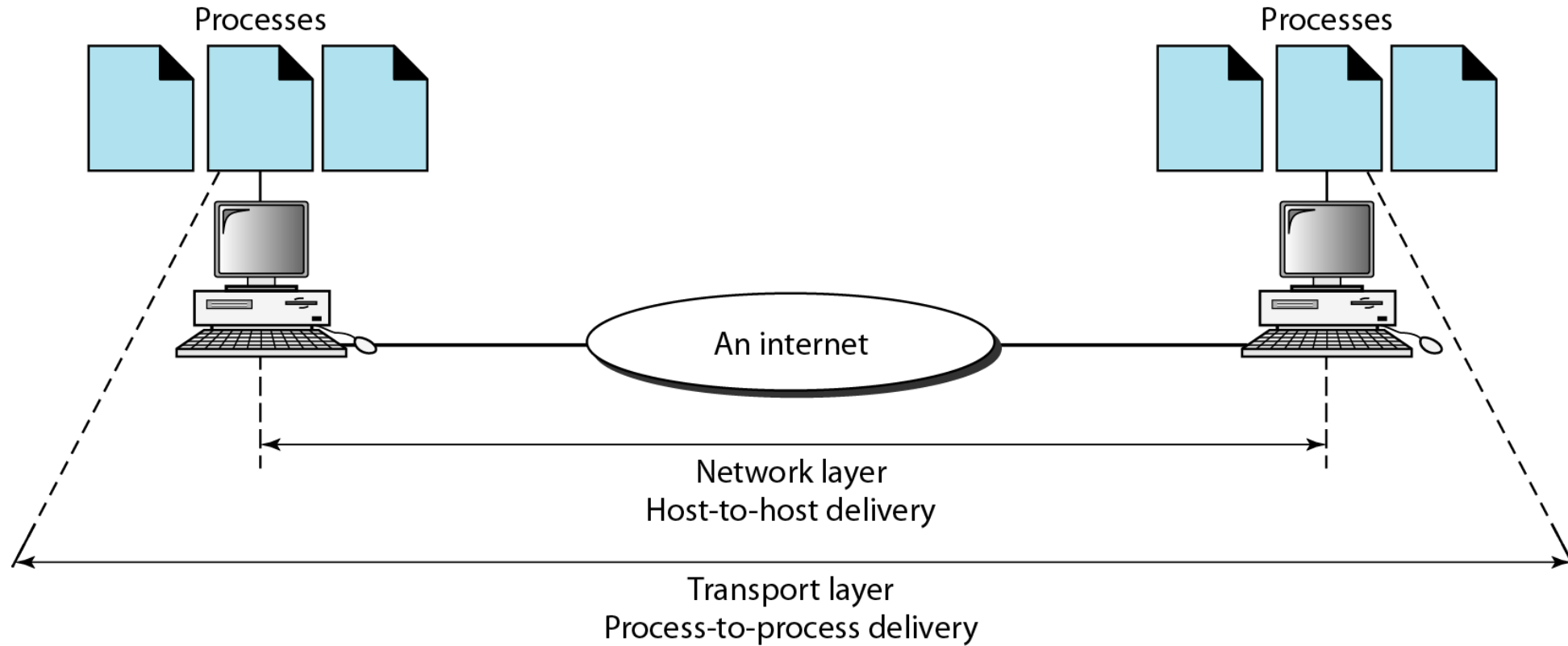
The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Transport Layer

- Makes sure data gets delivered to a specific process on a specific machine error free, in sequence, and with no losses or duplications.
- End-to-end protocol.
 - sender and receiver
- Handles retransmissions, if needed.
- Handles duplicates, if needed.
- Handles the FLOW CONTROL.
[A Fast Host cannot overrun the a slow one.]
- Also deals with addressing.
 - Which process on a particular machine?
- ***The transport layer is responsible for the delivery of a message from one process to another.***



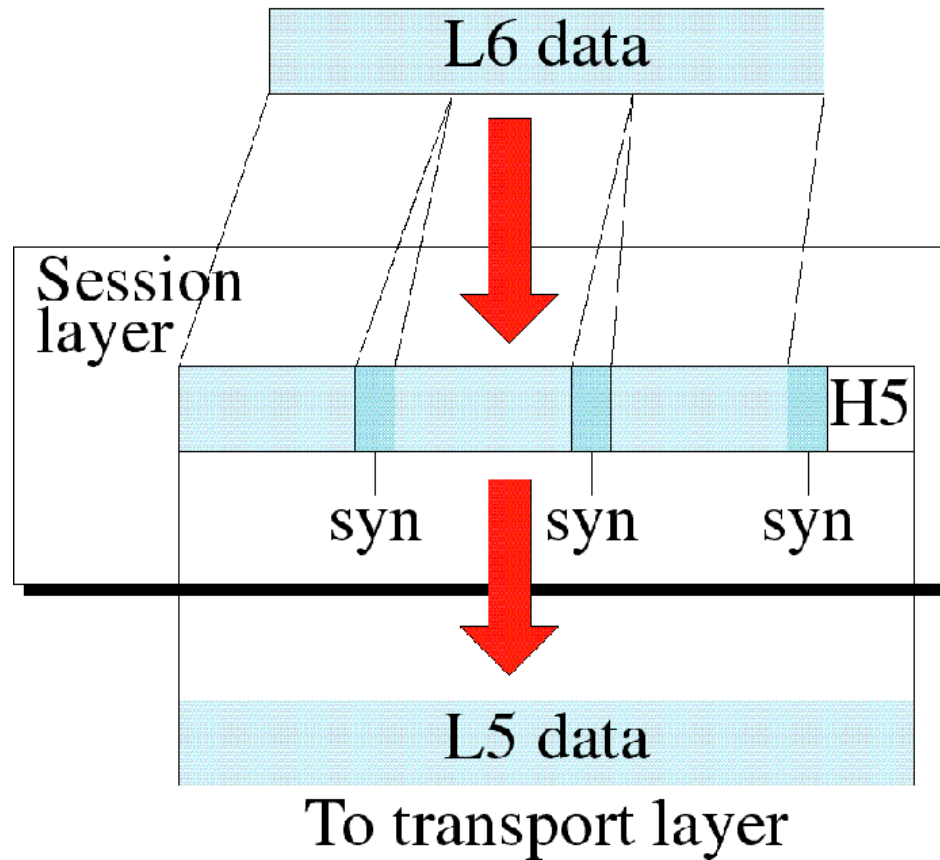
Reliable process-to-process delivery of a message



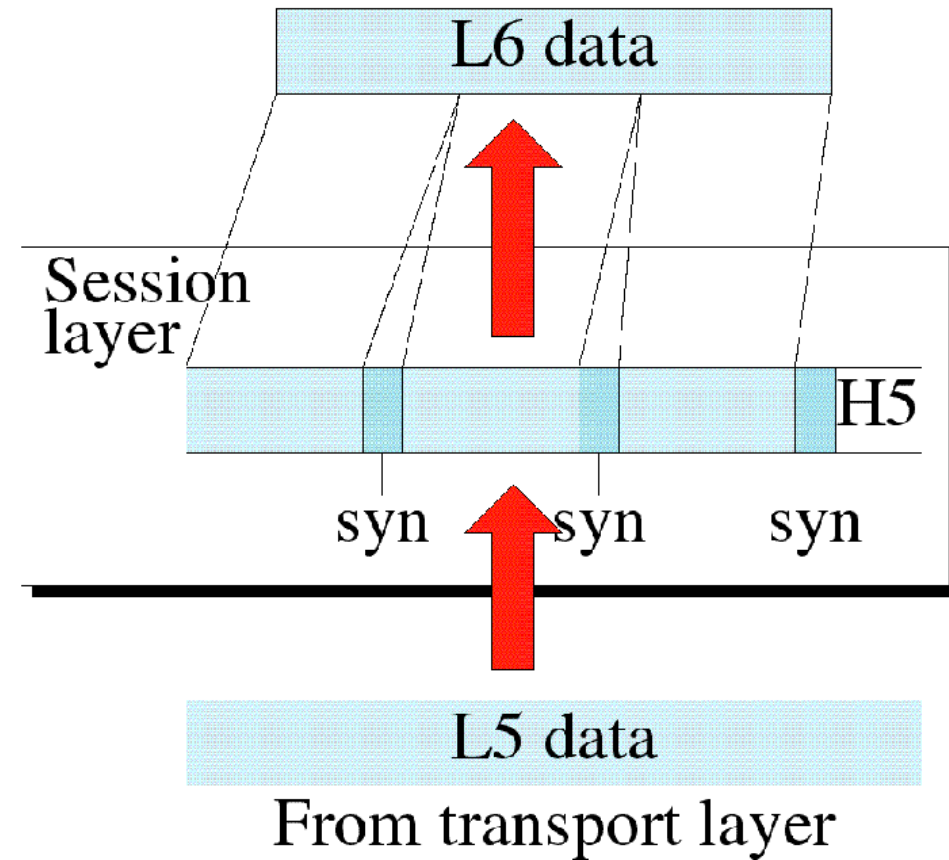
Session Layer

- Allows users to establish sessions between them.
- Provide services such as name lookup and security to allow two programs to find each other and establish the communications link.
- Manage dialogue control. [who can transmit & who can receive at what point during communication.]
- Provides synchronization.
 - Provides a way to insert checkpoints into the data stream , so that after the crash, only the data transferred after the last checkpoints have to be repeated.
- General services:
 - Compressing data.
 - Encryption.
- ***The session layer is responsible for dialog control and synchronization.***

From presentation layer

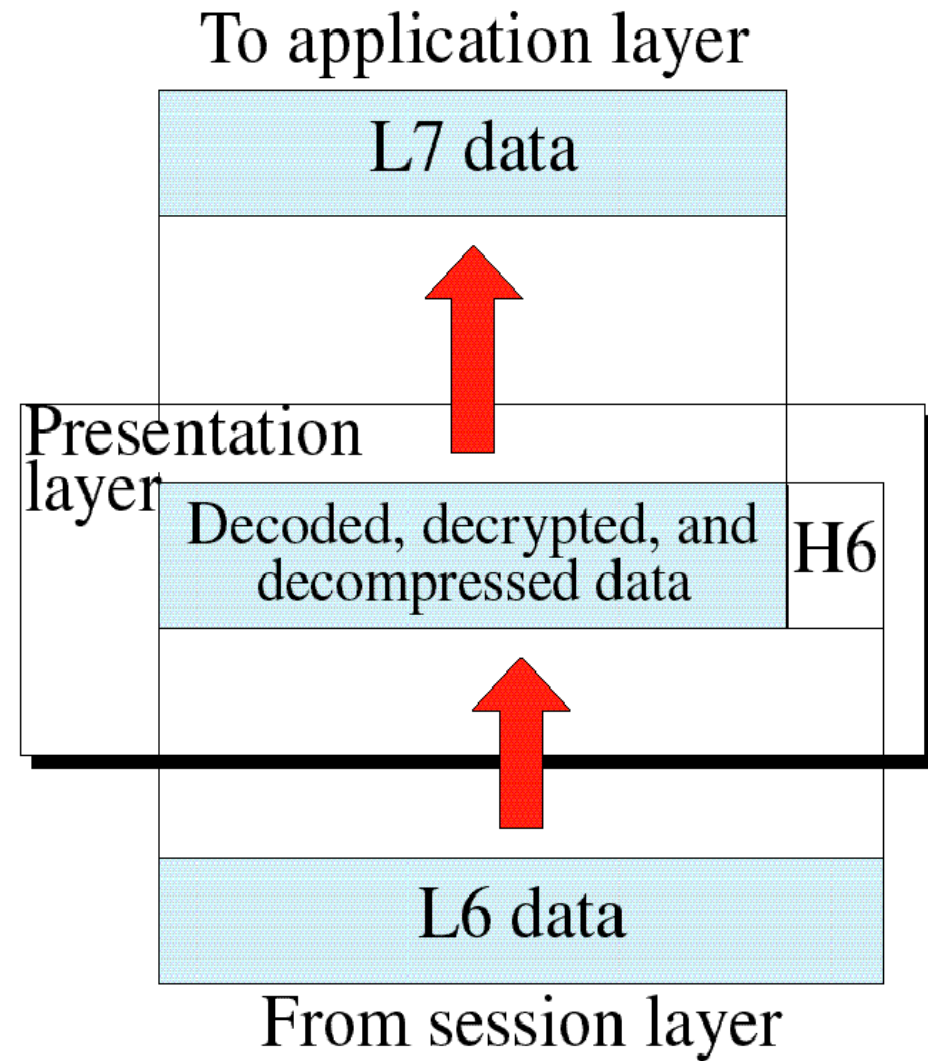
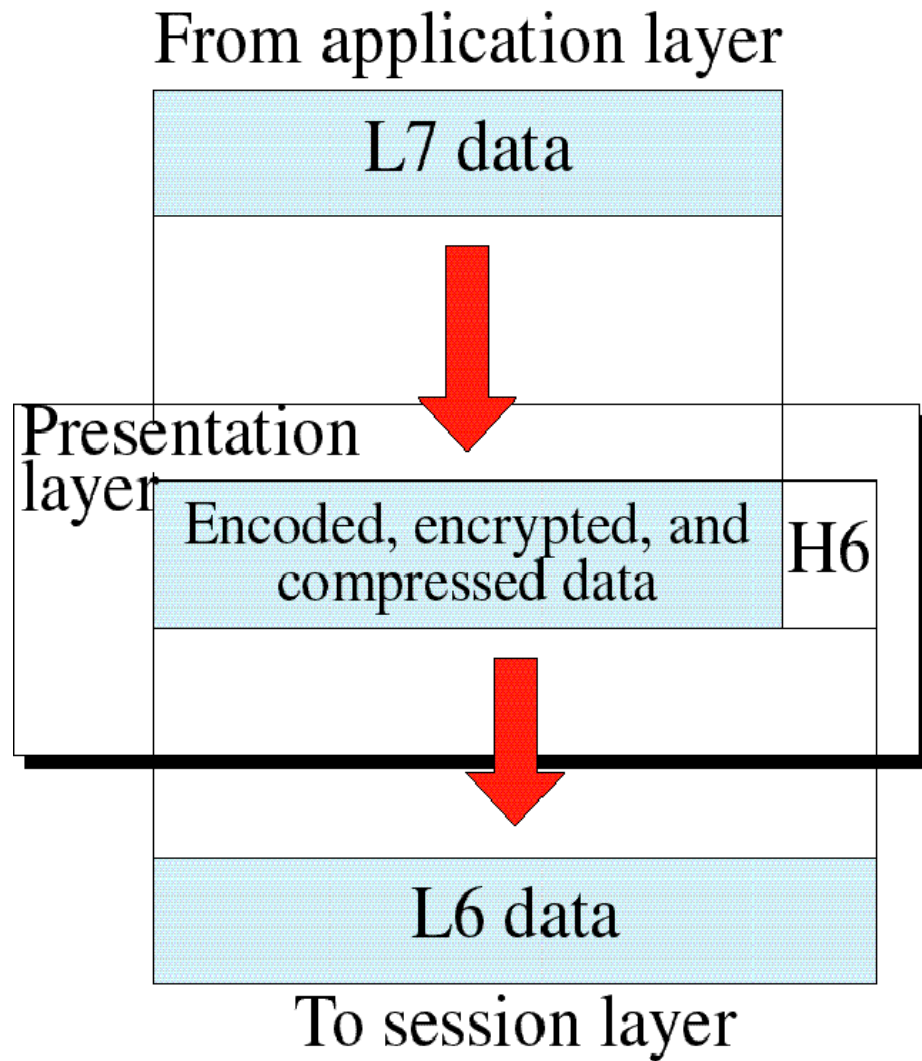


To presentation layer



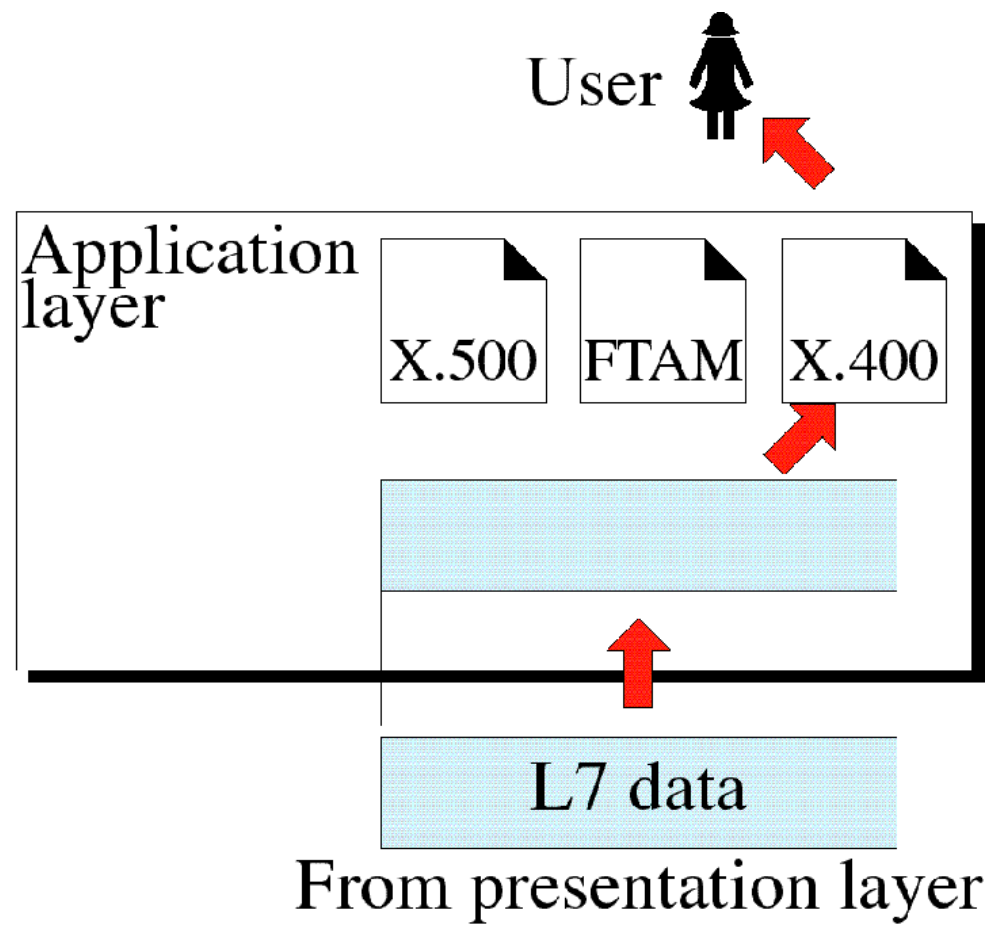
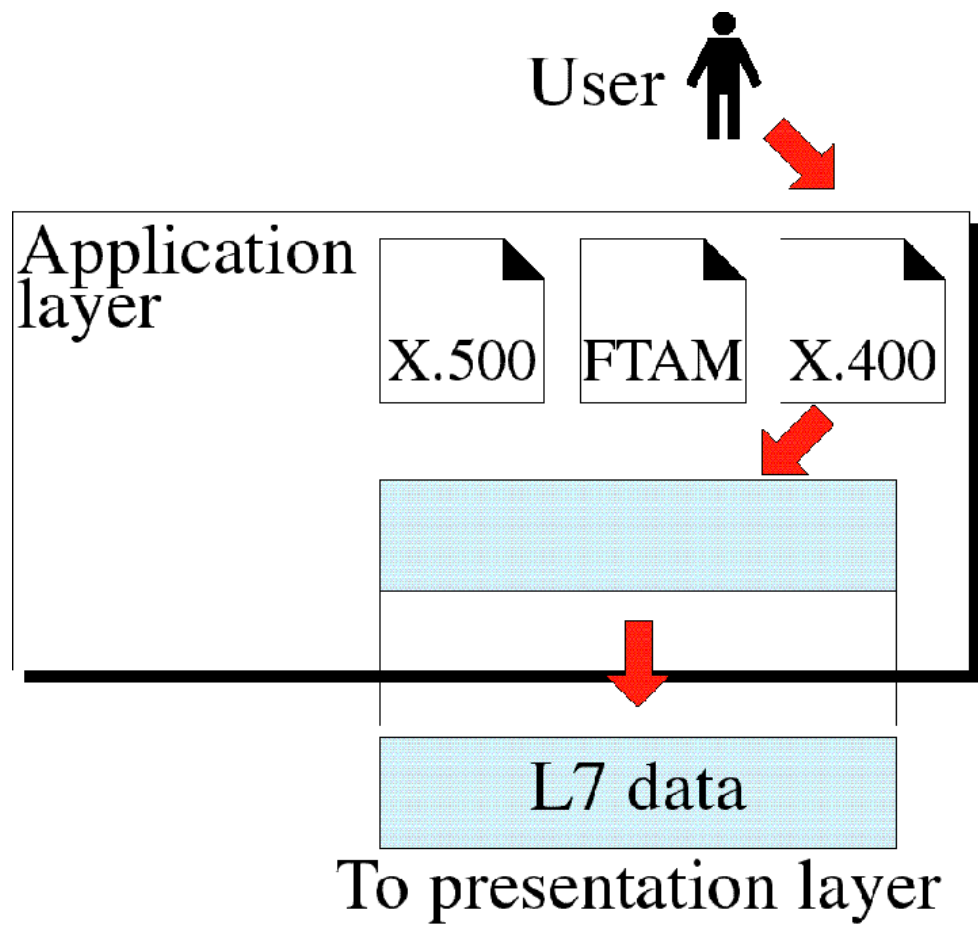
Presentation Layer

- Concerned with syntax & semantics of the information transmitted.
example: name, address ...
- Format in agreed upon way.
[In order to make it possible for computers with different representations to communicate, the data structure to be exchanged can be defined in an abstract way along with standard encoding.]
- *The presentation layer is responsible for translation, compression, and encryption.*

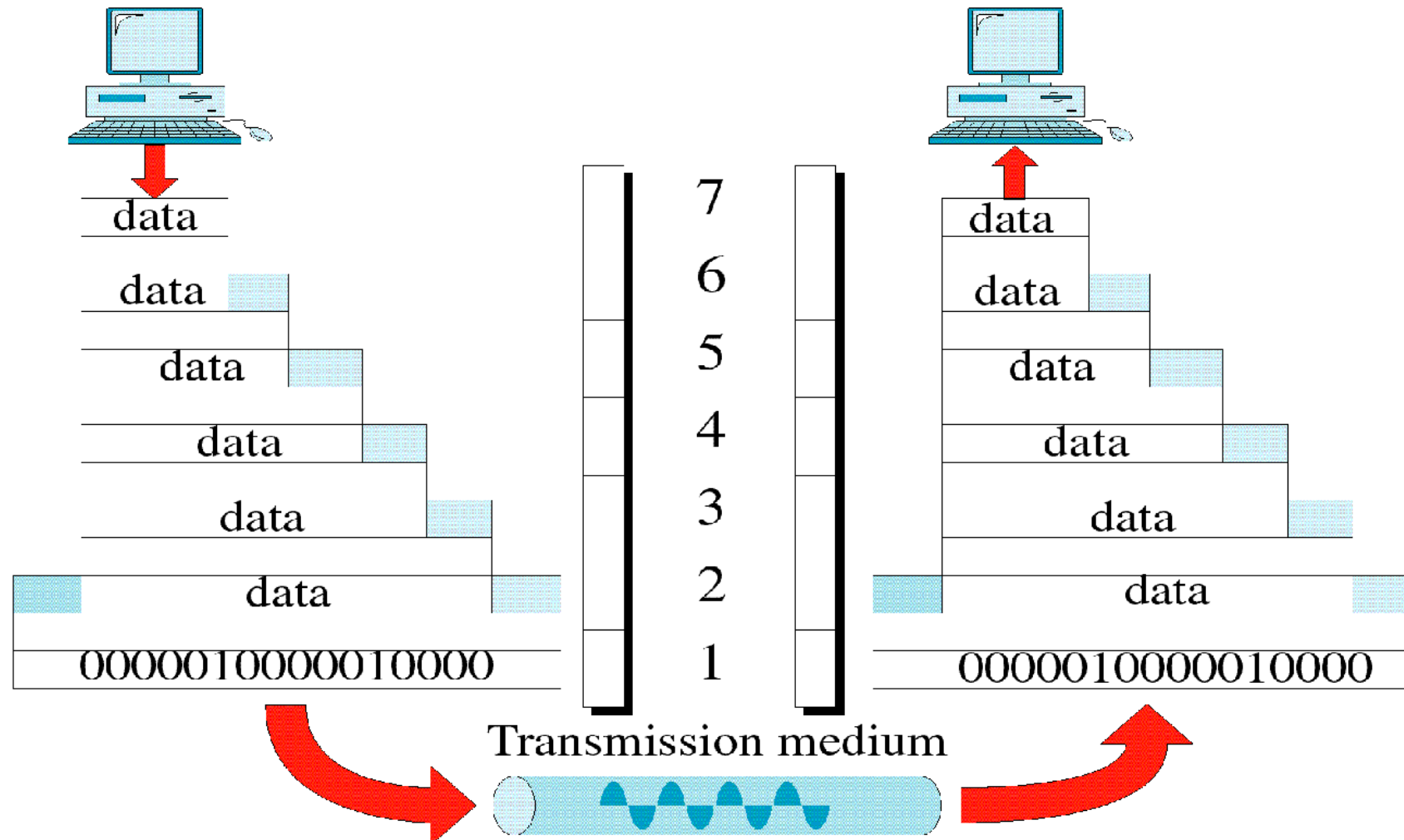


Application Layer

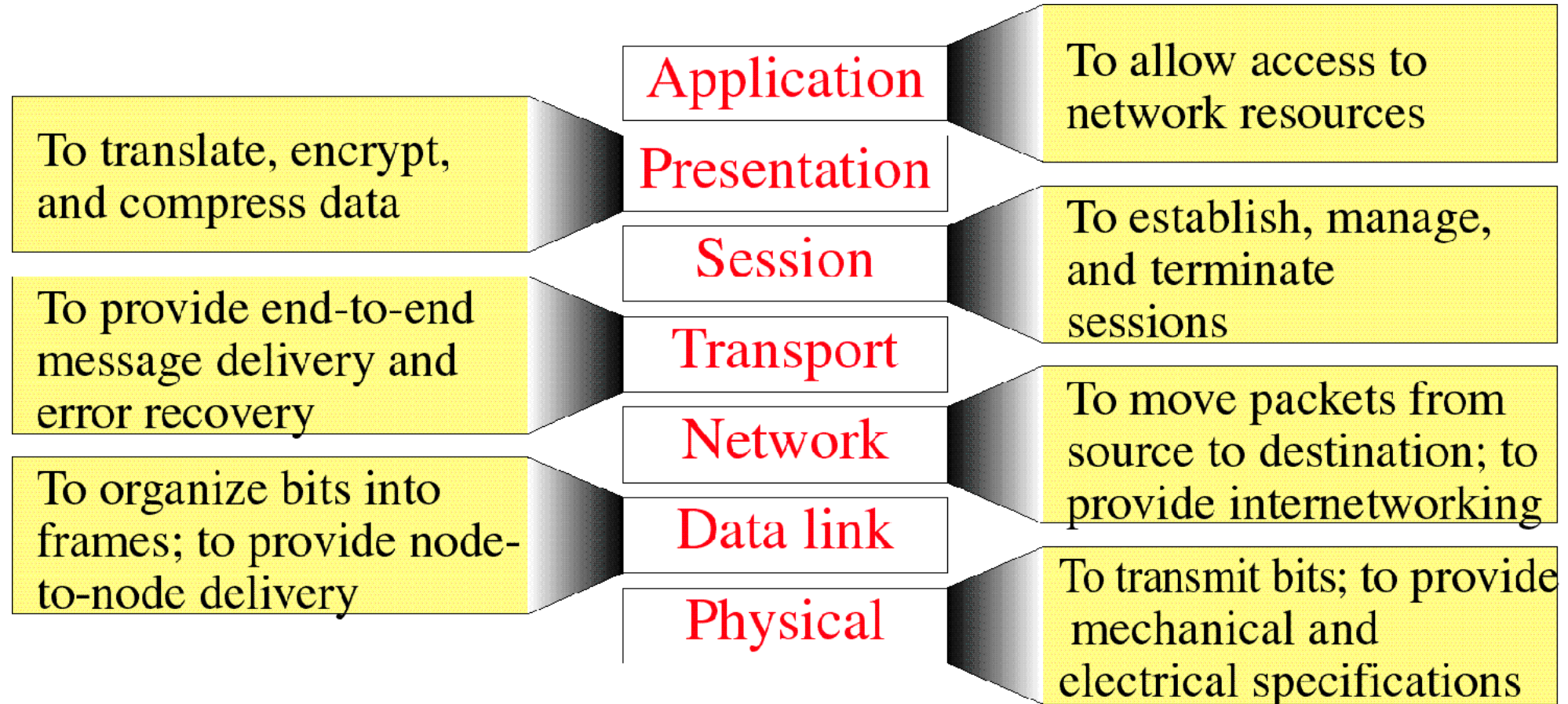
- It provides services that directly support user applications, such as database access, e-mail [Simple Mail Transfer Protocol-SMTP] , and File Transfer Protocol [FTP].
- It also allows applications to communicate with applications on other computers as though they were on the same computer.
- *The application layer is responsible for providing services to the user.*



An Exchange Using the OSI Model



Summary of Layer Functions



WHY SEVEN LAYERS :

1. A layer should be created where a different level of abstraction is needed.
2. Each layer should perform a well-defined function.
3. The function is chosen by considering the International standardized protocols
4. The Layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough so that distinct functions do not have to be combined into the same layer, and small enough that the architecture does not become too complex.

Advanced Research Projects Agency Network-

ARPANET

- Predecessor to Internet
- Phone lines first, satellite and radio later
 - req: connect multiple networks seamlessly
- Department of Defense-DoD worry about routers going down
 - req: survive loss of subnet hardware without losing connections
- Applications with diverse requirements
 - req: flexible architecture
- Used TCP/IP protocols
 - then came reference model

TCP/IP MODEL

- TCP/IP is the Transmission Control Protocol and the Internet Protocol, as well as a suite of related protocols developed by the Department of Defense's Advanced Projects Research Agency (ARPA; later DARPA) under its project on network interconnection, started in 1969.
- TCP/IP is by far the most widely used protocol for interconnecting computers, and it is the protocol of the Internet. ARPA originally created TCP/IP to connect military networks together, but it provided the protocol standards to government agencies and universities free of charge.

TCP/IP in Today's World

- Today, almost every Internet-enabled service communicates using the TCP/IP protocol suite, including:
 - ☐ Google Search
 - ☐ YouTube
 - ☐ WhatsApp
 - ☐ Amazon
 - ☐ AWS Cloud
 - ☐ Microsoft Azure
 - ☐ Netflix
 - ☐ UPI
 - ☐ IoT Devices
 - ☐ 5G Mobile Networks
- Without TCP/IP, these services would not be able to communicate across the Internet.

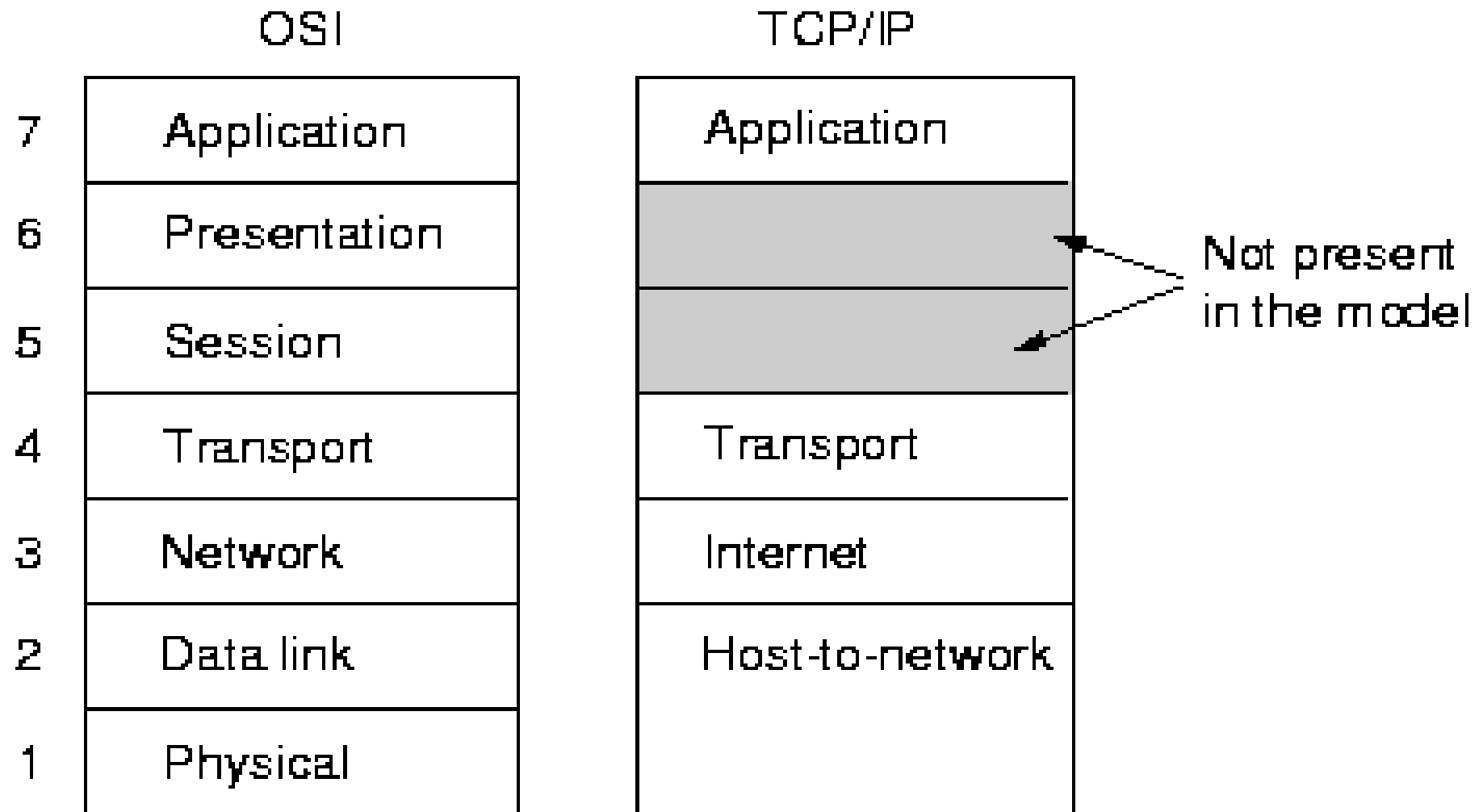


Fig. 1-18. The TCP/IP reference model.

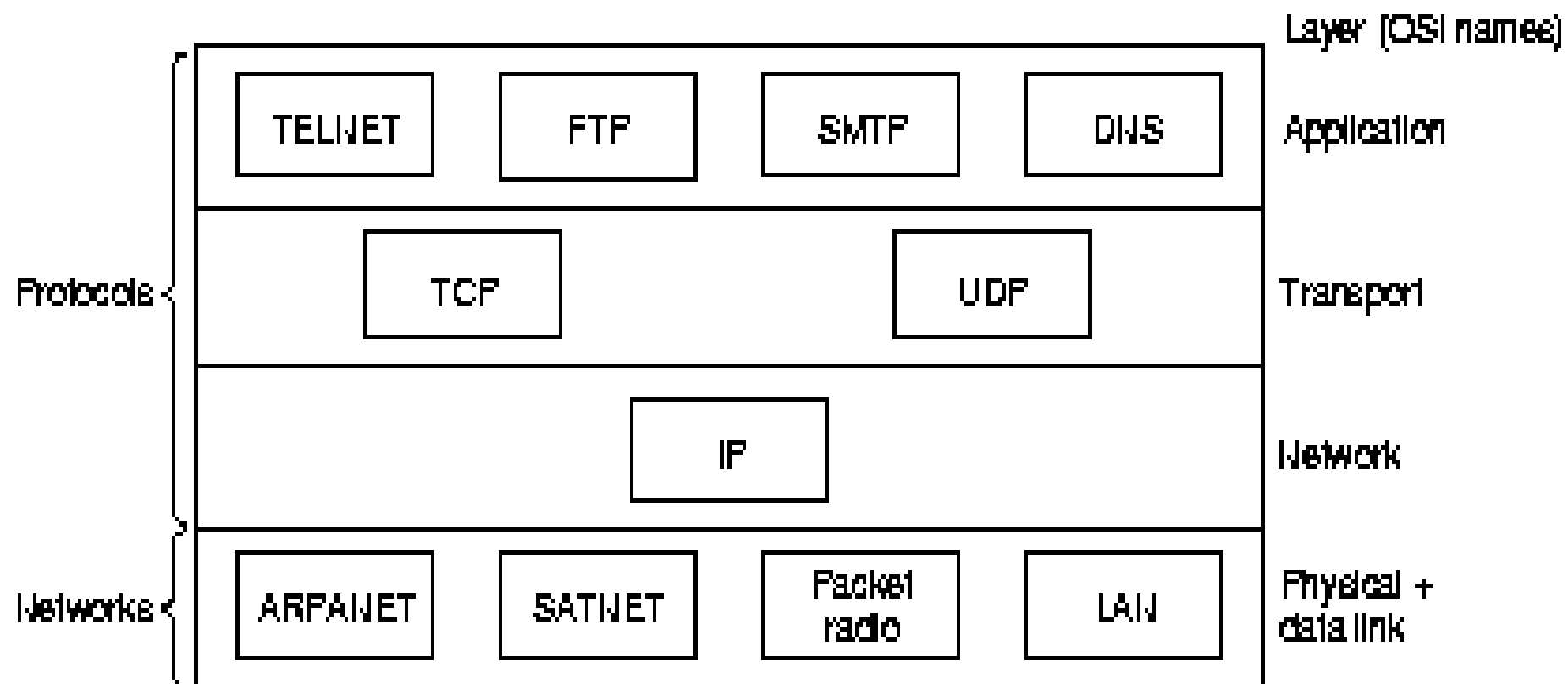
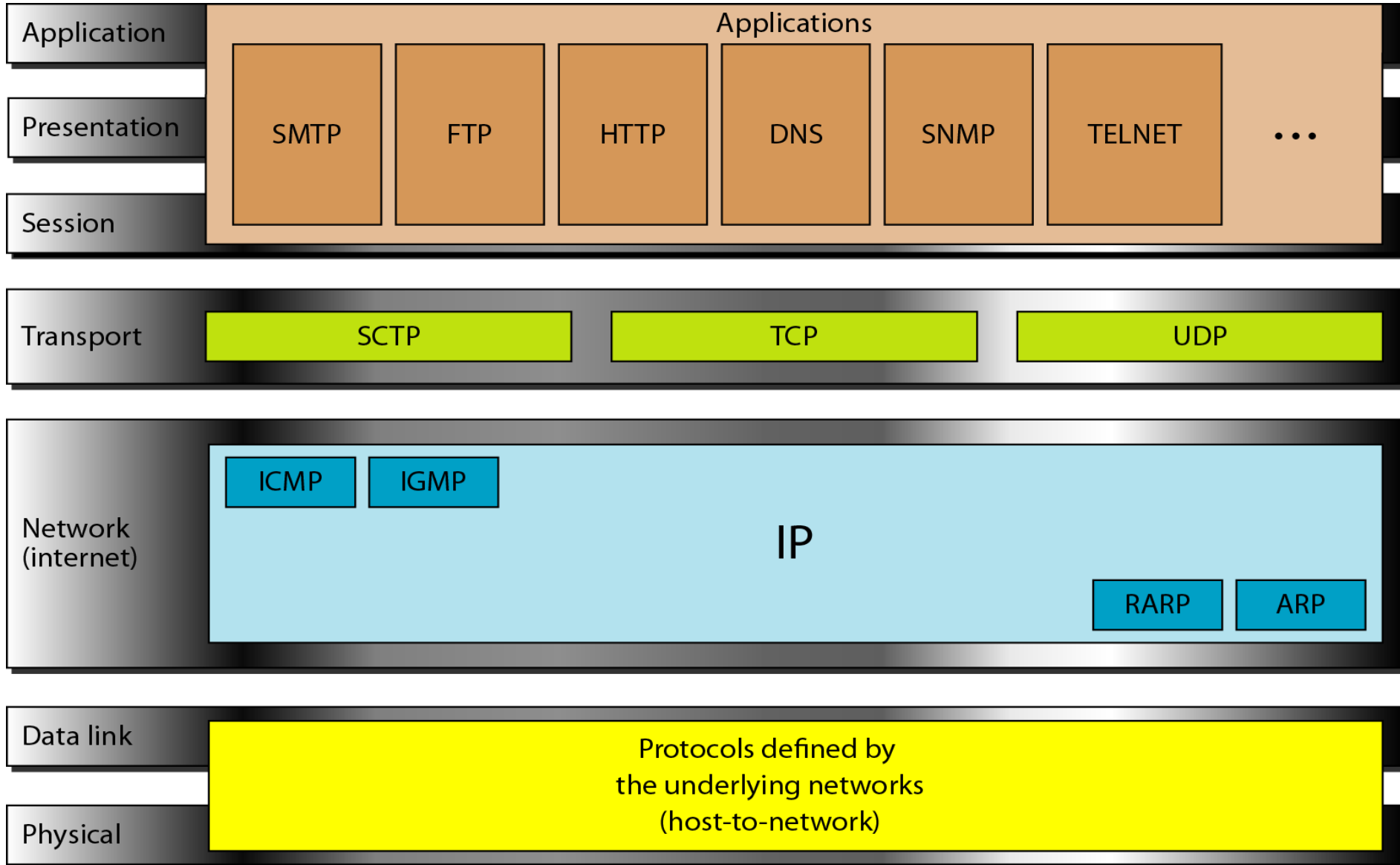


Fig. 1-19. Protocols and networks in the TCP/IP model initially.



Application Layer

- The application layer in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model.
- Contains higher-level protocols.
- TELNET- *virtual terminal protocol allows a user on one m/c, to log onto distant m/c & work there.*
- FTP-*provides a way to move data efficiently from one m/c to another.*
- DNS- *mapping host names onto their n/w addresses*
- HTTP- *protocol for fetching pages on the World Wide Web.*

TRANSPORT LAYER

- It allows peer entities on the source & the destination hosts to carry on a conversation, the same way as in the OSI transport layer.
- Two end-to-end protocols have been defined here.
- The first one, TCP (Transmission Control Protocol), is a reliable connection-oriented protocol that allows a byte stream originating on one machine to be delivered without error.
- It handles the FLOW CONTROL.
- The second protocol in this layer is UDP[User Datagram Protocol]. UDP is an unreliable , connectionless protocol for application that do not want TCP's sequencing or flow control and wish to provide their own.
- UDP is used for applications in which prompt delivery is more important than accurate delivery such as multimedia, speech etc.

- INTERNET LAYER:

- ❖ The Internet layer defines an official packet format and protocol called IP [Internet Protocol].
- ❖ The job of the Internet Layer is to deliver IP packets where they are required to go.
- ❖ Packet routing is clearly the major issue here, as is avoiding congestion. For this reason , we can say that TCP/IP internet layer is very similar in functionality to the OSI network layer.

- HOST-TO-NETWORK LAYER:

- ❖ This protocol is not defined and varies from host to host & network to network.

ACTIVITY: DRAG AND DROP THE EXAMPLES INTO THE CORRECT OSI LAYER



Drag and drop the
application-based examples
into the correct OSI layer.
(Up to Network Layer)



Web Browser
(HTTP)



Email
(SMTP)



File Transfer
(FTP)



WhatsApp
Messaging



Remote Login
(SSH)



Video Streaming
(YouTube)



DNS Lookup
(DNS)



IP Routing
(IP)

Application Layer (Layer 7)

Presentation Layer (Layer 6)

Session Layer (Layer 5)

Transport Layer (Layer 4)

Network Layer (Layer 3)

Feature	OSI Reference Model	TCP/IP Reference Model
Full Form	Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
Developed By	ISO (International Organization for Standardization)	DARPA (U.S. Department of Defense)
Year Developed	1984	1970s
Purpose	Standard reference model for network communication	Practical model for Internet communication
Number of Layers	7 Layers	4 Layers (sometimes shown as 5)
Nature	Theoretical (Conceptual)	Practical (Implementation-based)
Protocol Independent?	Yes, does not define specific protocols	No, based on TCP/IP protocol suite
Layer Separation	Clear separation of services, interfaces, and protocols	Less strict separation between layers
Reliability	Depends on the protocol used	Reliability provided mainly by TCP
Real-world Usage	Mainly for teaching and understanding networking concepts	Used on the Internet and modern computer networks
Flexibility	Less flexible	Highly flexible and scalable
Examples of Protocols	Does not specify protocols	HTTP, HTTPS, FTP, SMTP, DNS, TCP, UDP, IP, ICMP

Simple Analogy

OSI Model

- Think of the OSI model as the blueprint of a house.
- It explains how a house should be designed.
- It provides standards.
- It is mainly used for learning and planning.

TCP/IP Model

- Think of the TCP/IP model as the actual house that people live in.
- It is built according to practical needs.
- It is used every day.
- The Internet works using this model.

Insight

- Cisco routers, Juniper switches, Linux servers, and cloud platforms all communicate using standardized layered protocols.
- Network engineers troubleshoot issues layer by layer using the OSI model, even though the Internet runs on TCP/IP.